

PatientTriage.ai: Continuous Emergency Department Triage with Hybrid Deterministic Safety Rules and Advisory Machine Learning

Technical Submission & Architecture Specification Document

Format: Competition Technical README Document (PDF Export Ready)

System Status: Functional Decision-Support Prototype

Repository: https://github.com/sahilsheoran999/patient_triage

PROTOTYPE SIMULATION & REGULATORY NOTICE

PatientTriage.ai is a clinical decision-support prototype designed for emergency-department triage monitoring.

- **Synthetic Demonstration Data:** All patient cases, vital trends, and triage trajectories within this prototype are constructed from synthetic demonstration datasets.
- **No Autonomous Clinical Action:** The system does not diagnose conditions, prescribe medications, or replace clinical judgement.
- **Hierarchy of Authority:** Licensed clinicians retain 100% final decision authority over every triage classification. Deterministic safety rules act as an unbreachable safety floor that cannot be downgraded by the advisory machine learning model.
- **Regulatory Status:** This software is an engineering prototype and has not undergone prospective clinical trials, Institutional Review Board (IRB) review, or certification as Software as a Medical Device (SaMD) by CDSCO, US FDA, or European CE authorities.

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1. Executive Summary

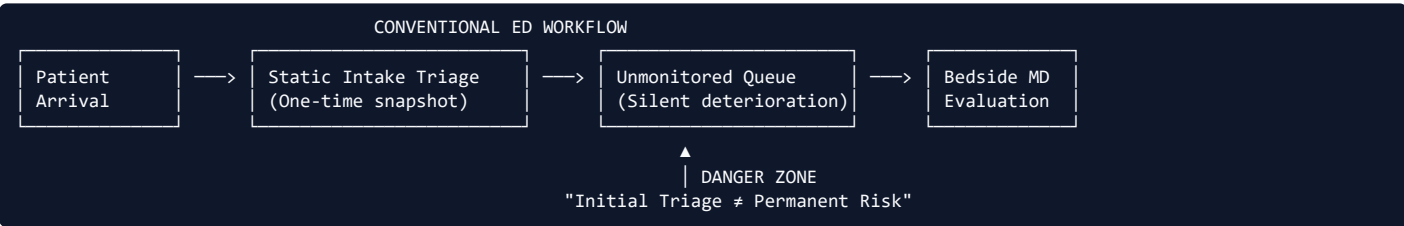
Emergency departments worldwide face severe overcrowding, volatile surge conditions, and acute clinician staffing shortages. Traditional triage processes classify patients once upon intake into static categories (such as ESI 1–5). However, a patient’s physiological state is dynamic. When waiting times stretch into hours, patients frequently experience silent clinical deterioration that goes unnoticed until a catastrophic event occurs.

PatientTriage.ai is an advanced clinical decision-support prototype engineered to address the emergency department waiting-room safety gap. The system introduces:

- 1. **Continuous Queue Surveillance (Waiting-Room Radar™)**: Shifts emergency triage from a single static snapshot at registration to continuous dynamic surveillance that tracks vital signs, trend trajectories, and elapsed wait times.
- 2. **Hybrid Safety Fusion Architecture**: Integrates hard deterministic safety rules (Safety Authority) with a 71-feature multi-class XGBoost model (Advisory Signal). Deterministic red flags enforce non-negotiable **CRITICAL** safety floors, preventing machine learning false-negative downgrades.
- 3. **UNKNOWN ≠ NORMAL Data Governance**: Missing clinical inputs (e.g., unmeasured blood pressure) are never defaulted to normal baselines. Missing data explicitly reduces data completeness, elevates system uncertainty, and triggers safety-first review policies.
- 4. **Transparent Explainability**: Features Tree SHAP (SHapley Additive exPlanations) attribution to surface exact physiological factors driving each advisory recommendation.
- 5. **Absolute Human Governance**: Licensed clinicians maintain final decision authority, backed by a structured override mechanism and an immutable event audit trail.

2. Problem Framing

In modern emergency departments, the triage desk serves as the critical gatekeeper. Triage nurses must rapidly assess complex patients—often with incomplete histories and unmeasured vital parameters—under intense time constraints.



Core Clinical & Operational Challenges:

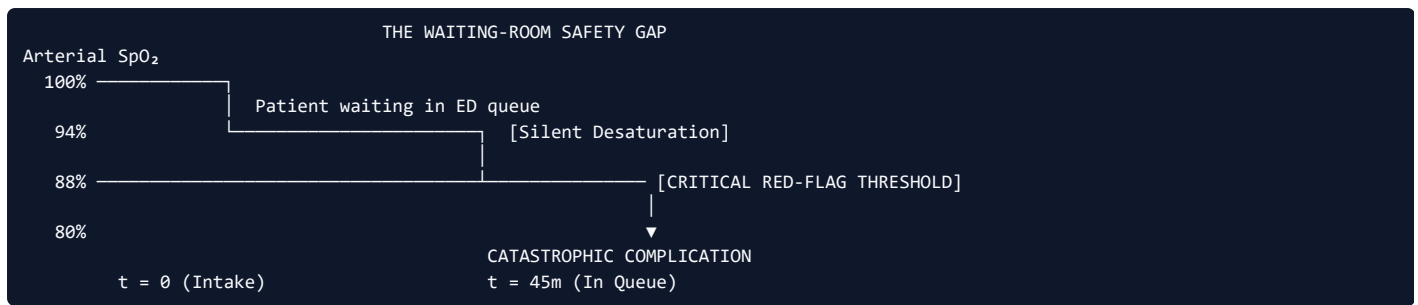
- 1. **Dynamic Physiological Drift**: Patients classified as moderate risk on arrival can deteriorate rapidly due to progressive sepsis, silent hypoxemia, or evolving myocardial ischemia.
- 2. **Cognitive Overload During Surges**: During high-volume surges, triage staff cannot manually reassess dozens of queued patients according to standard interval protocols.
- 3. **Missing Data Vulnerability**: Rushed triage frequently results in missing vitals. Conventional automated scoring algorithms often impute "normal" default values, masking severe physiological risk.
- 4. **AI "Black Box" Distrust**: Clinicians legitimately reject purely statistical or deep-learning triage algorithms that lack verifiable safety guarantees or can randomly recommend discharging an unstable patient.

3. Why Existing Triage Has a Safety Gap

Existing five-level triage systems (e.g., Emergency Severity Index / ESI, Manchester Triage System / MTS) rely on static heuristics executed at a single point in time:

$$\text{Patient Safety Risk} = f(\text{Vitals}(t_0), \text{Complaint}(t_0), \text{Elapsed Wait Time}(t - t_0), \Delta\text{Physiology}(t))$$

When systems ignore the time-dependent term $\Delta\text{Physiology}(t)$ and elapsed wait duration, serious safety failures emerge:



1. **Snapshot Fallacy:** A single normal vital reading at $t = 0$ provides zero guarantee of clinical stability at $t = 45m$.
2. **Silent Deterioration:** Chronic desaturation or progressive tachycardia often produces minimal outward distress until physiological compensation collapses.
3. **Threshold Blindness:** Patients assigned to lower acuity tiers frequently exceed standard waiting-time thresholds without triggering automated re-evaluation prompts.

4. Proposed Solution: Hybrid Decision Support

PatientTriage.ai addresses this challenge through a multi-tiered architecture that synthesizes deterministic clinical rules with machine learning advisory intelligence:

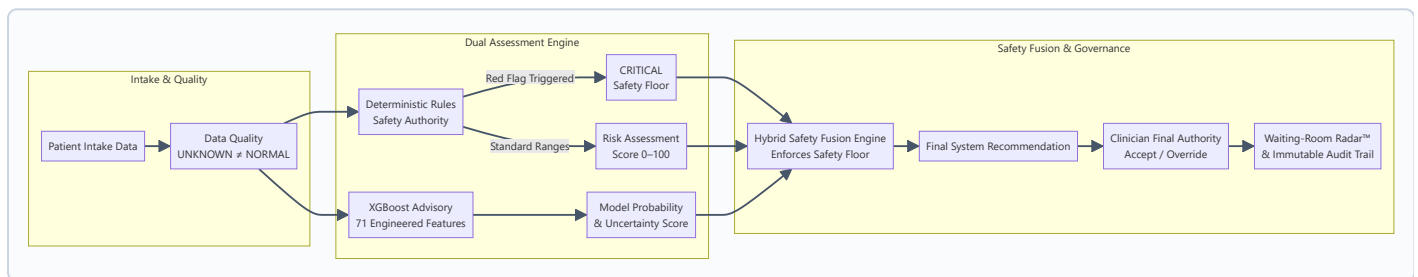


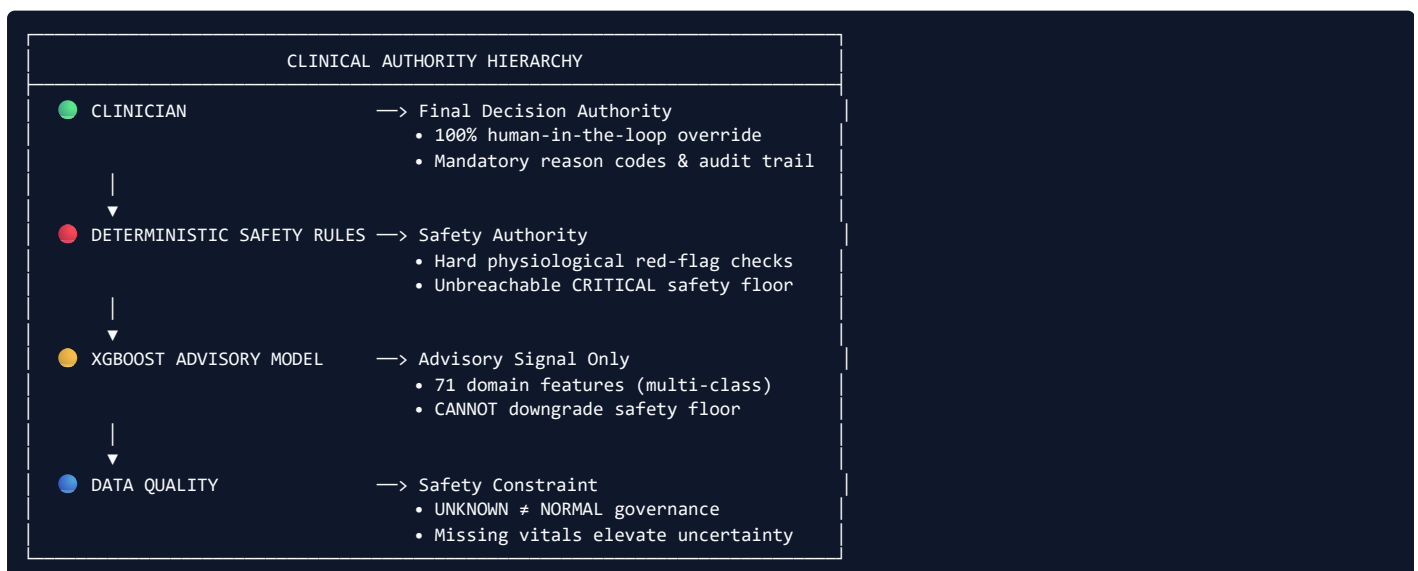
Figure 1 — End-to-End Hybrid Decision Support Architecture

Core Tenets of the Solution:

- **Safety Rules Have Authority Over ML:** Machine learning can upgrade acuity based on complex multi-variable interactions, but it can **never** downgrade a deterministic safety floor.
- **Continuous Active Surveillance:** The Waiting-Room Radar™ evaluates queued patients on every vital update and clock tick, raising contextual "Why Now?" alerts when deterioration occurs.
- **Explainability at the Point of Care:** Every advisory score displays exact Tree SHAP mathematical attributions alongside a transparent factor decomposition.

5. System Architecture

Clinical Authority Hierarchy



PatientTriage.ai implements an 11-layer sequential pipeline separating data ingestion, safety constraints, statistical advisory inference, human control, and longitudinal surveillance:

PATIENTTRIAGE.AI 11-LAYER ARCHITECTURE		
Layer	Component Name	Functional Responsibilities
L-01	Patient Data Ingestion	Captures vitals, complaints, history, cues
L-02	Data Quality & Completeness	UNKNOWN ≠ NORMAL: tracks missingness flags
L-03	Age-Aware Normalization	Pediatric, Adult, and Geriatric baselines
L-04	Deterministic Safety Engine	Evaluates hard red flags (Safety Authority)
L-05	Rule-Based Risk Assessment	Computes transparent weighted risk (0-100)
L-06	XGBoost Advisory Model	71-feature multi-class inference ensemble
L-07	Probability & Uncertainty	Softmax class distributions & margin score
L-08	Hybrid Safety Fusion	Enforces safety floor over ML predictions
L-09	Final System Recommendation	Synthesized decision-support output
L-10	Clinician Review Authority	Final human decision (Accept/Override)
L-11	Audit & Continuous Radar	Immutable logging + Waiting-Room Radar™

6. End-to-End Clinical Decision Flow

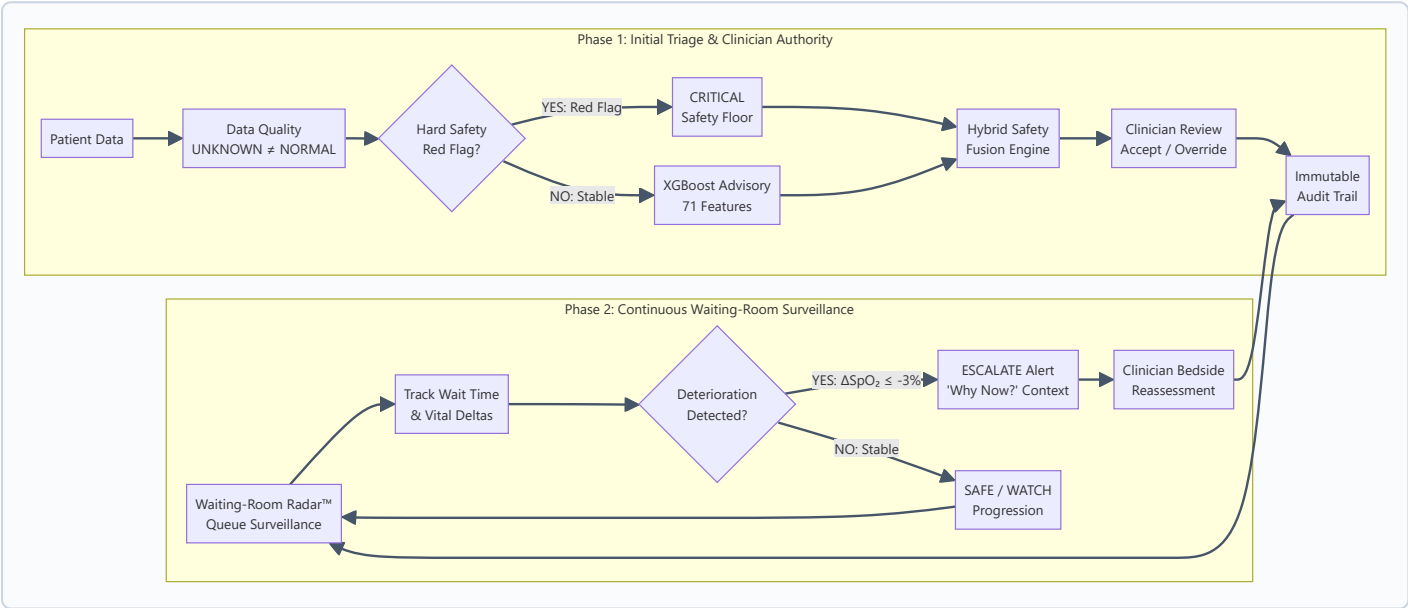


Figure 2 — End-to-End Clinical Decision and Continuous Monitoring Flow

The architecture separates initial triage from continuous monitoring. Deterministic safety rules establish non-downgradable safety floors, while XGBoost provides an advisory probability distribution. The clinician retains final decision authority. After registration, Waiting-Room Radar continuously evaluates elapsed wait time and vital-sign trends and can trigger reassessment when predefined prototype conditions are met.

7. Deterministic Safety Layer (Safety Authority)

The Deterministic Safety Layer operates as the primary safety gatekeeper. If a patient exhibits objective physiological instability matching predefined red-flag criteria, the system asserts an immediate **CRITICAL** safety floor.

Hard Physiological Red-Flag Thresholds

SafetyFloor = {

- CRITICAL, if SpO₂ < 88%
- CRITICAL, if SBP < 80 mmHg or ≥ 200 mmHg
- CRITICAL, if RR ≥ 30 /min
- CRITICAL, if HR ≥ 140 bpm or ≤ 40 bpm
- CRITICAL, if Temp ≥ 40.5°C or ≤ 35.0°C
- CRITICAL, if DistressCue ∈ {Stridor, Cyanosis, Confusion}
- PASS, otherwise

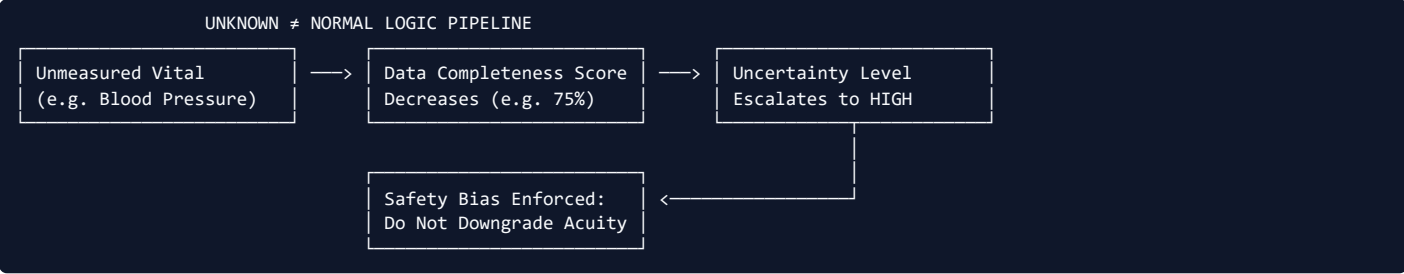
Age-Aware Baselines & Multipliers

The system adjusts normal ranges and scoring weights across three demographic cohorts:

Demographic Cohort	Age Bracket	Normal HR	Normal RR	Normal SBP	Risk Multiplier
Pediatric	< 18 years	70–130 bpm	20–30 /min	90–115 mmHg	1.25×
Adult	18–64 years	60–100 bpm	12–20 /min	100–130 mmHg	1.00×
Geriatric	≥ 65 years	55–95 bpm	14–22 /min	110–140 mmHg	1.30×

8. Data Quality & UNKNOWN ≠ NORMAL Principle

A foundational safety innovation in PatientTriage.ai is explicit handling of missing data.



Technical Implementation:

- No Silent Normal Defaults:** If the intake nurse leaves systolic blood pressure blank, it is stored as `null`. It is **never** coerced to 120 mmHg.
- Completeness Formula:**
$$\text{Completeness} = \left(\frac{\sum_{i=1}^9 \mathbb{I}(\text{field}_i \neq \text{null})}{9} \right) \times 100\%$$
Fields evaluated: SpO₂, HR, SBP, RR, Temp, ChiefComplaint, AssociatedSymptoms, MedicalHistory, Allergies.
- Safety Bias Under Uncertainty:**
 - If Uncertainty = HIGH and calculated priority is LOW or NON_URGENT, the system automatically escalates to MEDIUM.
 - If Uncertainty = HIGH and calculated priority is MEDIUM, the system escalates to HIGH.

9. Rule-Based Risk Assessment Layer

The Rule-Based Risk Assessment Layer calculates a transparent, fully decomposed numerical risk score (0–100) based on configured prototype weights:

$$\text{RawScore} = \sum w_i \cdot x_i$$

Prototype Weight Breakdown (`src/config/riskWeights.ts`)

Clinical Parameter	Prototype Weight	Description & Conditions
SpO ₂ Abnormality	32 points	< 88% → 32 pts; 88–92% → 24 pts; 93–94% → 13 pts; Missing → 12 pts penalty
Respiratory Symptoms	24 points	RR ≥ 26 /min → 24 pts; Acute dyspnea in chief complaint → 17 pts
Observed Clinical Distress	12 points	min(18, N _{cues} × 12 pts)
Age Risk Factor	10 points	10 pts × CohortMultiplier (Pediatric: 13, Geriatric: 13)
Heart Rate Abnormality	8 points	HR > 110 bpm or < 50 bpm
Blood Pressure Abnormality	6 points	SBP > 160 mmHg or < 90 mmHg
Temperature Abnormality	6 points	Temp ≥ 38.5°C or < 36.0°C
Medical History Comorbidity	4 points	min(16, N _{history} × 4 pts); Zero History Record → 14 pts penalty

Score to Priority Mapping:

- Score ≥ 90 ⇒ CRITICAL
- Score ≥ 70 ⇒ HIGH
- Score ≥ 45 ⇒ MEDIUM
- Score ≥ 25 ⇒ LOW
- Score < 25 ⇒ NON_URGENT

10. XGBoost Machine Learning Advisory Layer

The machine learning layer captures complex multi-variable interactions that simple linear rule sets may overlook.

Architecture Specifications

- **Classifier Type:** Multi-Class Gradient Boosted Decision Tree (XGBClassifier)
- **Objective Function:** multi:softprob across 5 acuity classes
- **Tree Structure:** 300 compact trees (maximum depth = 5, learning rate = 0.04, subsample = 0.85)
- **Input Dimension:** 71 engineered clinical features

71 Engineered Features → Tree Traversal (300 Trees) → Raw Margins (Logits) → Softmax Normalization → 5-Class Probabilities

Softmax Output Formulation

$$P(\text{Class } k \mid \mathbf{x}) = \frac{\exp(z_k)}{\sum_{j=1}^5 \exp(z_j)}, \quad z_k = \text{BaseMargin}_k + \sum_{m \in \text{Trees}_k} f_m(\mathbf{x})$$

11. Hybrid Safety Fusion Engine

The Hybrid Safety Fusion Engine reconciles deterministic rule evaluations with advisory machine learning predictions according to strict safety invariants.

Fusion Policies & Precedence Hierarchy

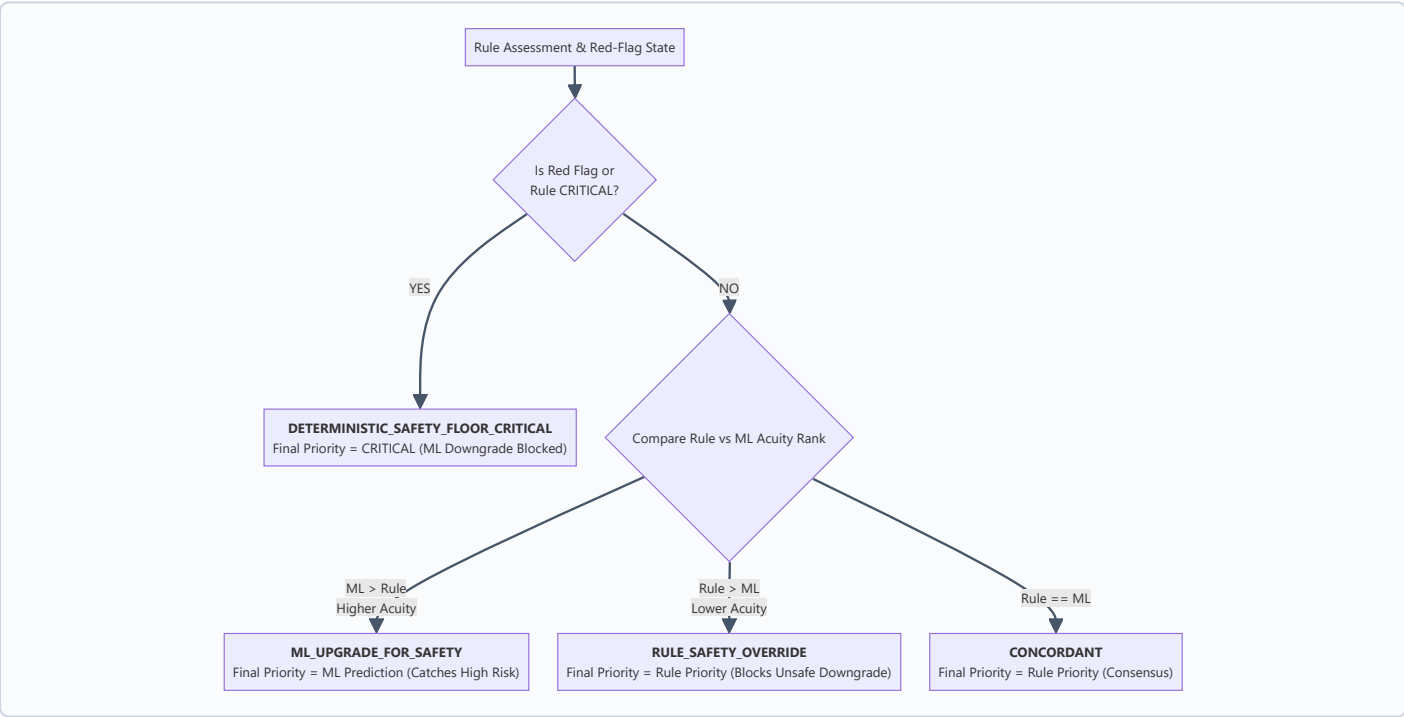


Figure 3 — Hybrid Safety Fusion Authority Precedence

Policy Definitions:

1. **DETERMINISTIC_SAFETY_FLOOR_CRITICAL** : Strictly enforces **CRITICAL** priority if red flags are active, regardless of ML output.
2. **ML_UPGRADE_FOR_SAFETY** : Accepts higher acuity ML predictions (e.g., rule **LOW** → ML **MEDIUM**) to catch multi-symptom presentations.
3. **RULE_SAFETY_OVERRIDE** : Rejects ML downgrade suggestions below rule-assessed priority.
4. **CONCORDANT** : Both engines agree on acuity.
5. **FALLBACK_DETERMINISTIC_ONLY** : Used if ML model inference fails or is disabled.

12. Model Explainability & SHAP Attribution

To provide clinicians with interpretable reasoning, the system calculates local feature attributions via Tree SHAP:

LOCAL SHAP ATTRIBUTION PANEL			
Clinical Feature	Value	Importance	Impact on Risk
Oxygen Saturation (SpO ₂)	84.0%	0.088	↑ Increases Risk
Systolic Blood Pressure (SBP)	78.0	0.082	↑ Increases Risk
Heart Rate	138	0.076	↑ Increases Risk
Chest Pain / Angina	Yes	0.061	↑ Increases Risk
Shock Index (HR/SBP)	1.77	0.054	↑ Increases Risk

Mathematical Interpretation Guardrail: SHAP values indicate mathematical feature importance within the decision forest. They represent statistical attribution rather than clinical causality.

13. Continuous Waiting-Room Monitoring (Waiting-Room Radar™)

Triage is an ongoing longitudinal process. The Waiting-Room Radar™ continuously evaluates all patients in the queue:

[Arrival] → [Initial Triage] → [Waiting Room Queue] → [Radar Surveillance Engine] → [Active Alerts]

The 4 Radar Monitoring States

State	Badge	Criteria	System Response
ESCALATE	Red	Physiological deterioration detected or safety red flag triggered	Loud UI alert, top-of-queue pinning, immediate bedside call
REASSESS	Orange	Elapsed wait time exceeds configured threshold	Reassessment timer banner, vitals refresh prompt
WATCH	Yellow	Active surveillance (high uncertainty or ML disagreement)	Passive highlighted card, elevated surveillance
SAFE	Green	Vitals stable, wait time within normal limits	Standard queue progression

14. Reassessment & Deterioration Detection Logic

Deterioration Detection Rules

The system evaluates sequential vital readings (vitalsHistory) and triggers deterioration alerts when:

$$\text{Deterioration} = (\Delta\text{SpO}_2 \leq -3\%) \vee (\Delta\text{HR} \geq +15 \text{ bpm}) \vee (\Delta\text{RR} \geq +6 / \text{min})$$

Contextual "Why Now?" Alerts

The Radar generates specific, truthful reasons explaining why an alert is triggering at this exact moment:

- SAFETY_RED_FLAG** : "Deterministic red flag detected: Critical Hypoxemia (SpO₂ 84% < 88%). Safety floor enforced."
- DETERIORATION** : "Vital trend deterioration: SpO₂ ↓ 7%; Heart Rate ↑ 23 bpm recorded 2m ago."
- MODEL_RULE_DISAGREEMENT** : "Deterministic rules classify patient as NON_URGENT (Score 22/100), while XGBoost predicts MEDIUM (81.9%). Clinician review recommended."
- WAIT_TIME_EXCEEDED** : "Patient has waited 82 minutes; configured MEDIUM-priority threshold is 30 minutes."
- HIGH_UNCERTAINTY** : "High uncertainty review: Missing blood pressure and medical history."

15. Clinician-in-the-Loop Governance

PatientTriage.ai is strictly engineered as decision support. The human clinician retains complete final authority.

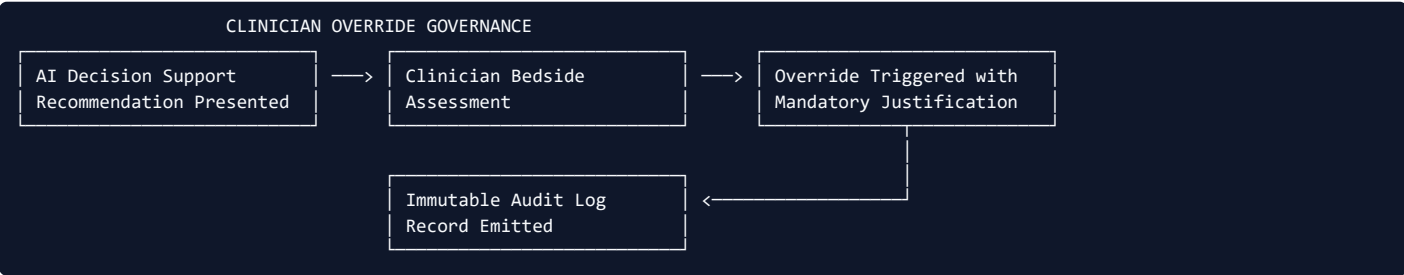


Figure 4 — Clinician Override and Audit Governance Flow

Clinician Rights & System Invariants:

- 1. Clinicians can override any AI recommendation to any priority level.
- 2. Every override mandates selecting a structured reason category and providing clinical notes.
- 3. Overrides are permanently recorded to the audit trail for quality assurance and model governance.

16. Audit Trail & Override Architecture

Every clinical event is emitted as an immutable structured log record (AuditLogEntry):

```
{
  "id": "LOG-1042",
  "timestamp": "2026-08-30T10:15:32Z",
  "patientId": "P-111",
  "patientName": "Deepak Verma",
  "user": "Dr. Neha Verma (MD)",
  "eventType": "CLINICIAN_OVERRIDE",
  "details": "Priority changed from MEDIUM to HIGH. Reason: New clinical observation (angina history; worsening diaphoresis).",
  "previousState": "MEDIUM",
  "newState": "HIGH",
  "severity": "alert"
}
```

17. User Interface & Mission Control Experience

The frontend is designed around dark clinical mission-control principles to maximize readability and minimize cognitive fatigue.

Core Views:

- **Command Center:** Real-time ED queue with 7 top KPI cards, search/filter controls, and multi-stat patient cards.
- **Waiting-Room Radar™:** Longitudinal timeline view tracking patient progression and active alerts.
- **Patient Intake:** Fast-entry form with real-time completeness gauge and explicit unmeasured data handling.
- **Patient Detail Modal:** Comprehensive 4-stat bar, vitals trend line chart, AI decision panel, and SHAP attribution bars.
- **Clinician Override Modal:** Structured 3-step override dialogue.
- **Safety Policy View:** Interactive 10-stage architecture diagram and policy documentation.
- **Hospital Configuration:** Multi-facility customization panel.

18. Dataset Generation & Training Methodology

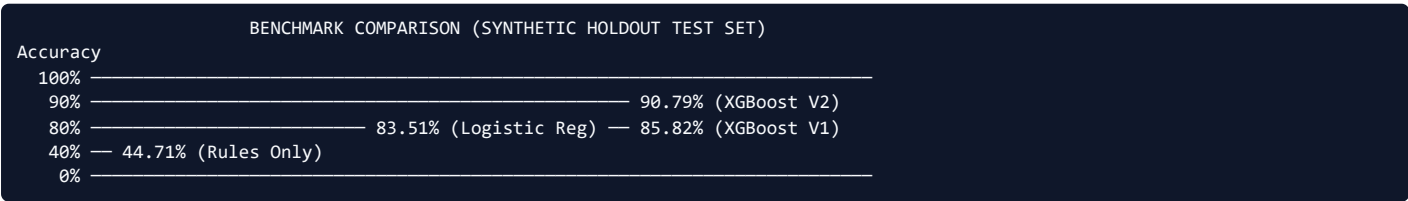
Synthetic Dataset Architecture

The training corpus consists of 15,000 synthetic clinical records generated using physiological distributions modeled after emergency department presentations:

- **Acuity Class Balance:** Stratified across CRITICAL , HIGH , MEDIUM , LOW , and NON_URGENT .
- **Augmentation:** Augmented with 1,000 conflicting-signal validation archetypes (e.g., severe hypoxemia with benign chief complaints; atypical coronary presentations with normal baseline vitals).
- **Partitioning:** Stratified 70% Training (10,500 records), 15% Validation (2,250 records), and 15% Holdout Test (2,250 records).

19. Model Evaluation & Synthetic Benchmark Results

Synthetic Holdout Benchmark Notice: The reported benchmark metrics reflect prototype performance on synthetic evaluation sets. They demonstrate architecture behavior and do not establish clinical safety or efficacy.



Comprehensive Benchmark Table

Model / Architecture	Accuracy	Macro F1-Score	Critical Class Recall	High + Critical Sensitivity
Deterministic Rules Only	44.71%	30.90%	91.53%	88.24%
Logistic Regression Baseline	83.51%	81.00%	92.82%	95.12%
XGBoost Advisory Model (V1)	85.82%	83.47%	95.21%	97.65%
XGBoost Domain-Robust Model (V2)	90.79%	88.20%	95.99%	98.87%

- **Zero Critical→Low/Non-Urgent Misclassifications in the Synthetic Holdout Test Set:** On the evaluated synthetic test partition (2, 250 records), 0 true Critical cases were classified into Low or Non-Urgent categories. *(Note: This result applies strictly to the evaluated synthetic test set and does not establish real-world clinical safety or diagnostic accuracy.)*
- **Exploratory Generalization Benchmark on De-identified EHR Sample:** Evaluated on an untouched holdout sample of 207 labeled records from the MIMIC-IV-ED database (`m1/data/triage.csv.gz` , ESI levels 1–4), the domain-robust feature extraction identified 88.89% (16/18) of Critical resuscitation cases (ESI Level 1) and 61.74% (71/115) of High+Critical cases (ESI Levels 1–2) in zero-shot evaluation without fine-tuning. This exploratory evaluation demonstrates prototype transferability under missing data conditions, but does not constitute prospective clinical validation.

20. Edge-Case Validation Scenarios

Case 1: P-127 (Atypical Presentation — ML Catches Interaction)

- **Presentation:** 45-year-old male with substernal chest pressure, vomiting, dizziness, Levine sign, and unmeasured blood pressure.
- **Rule Engine:** Scored 22/100 → NON_URGENT (missing vitals, benign keywords).
- **XGBoost Model:** Predicts MEDIUM (81.9% Model Probability) via non-linear cardiac interaction capture.
- **Safety Fusion:** Applies ML_UPGRADE_FOR_SAFETY → Final: MEDIUM (Disagreement flagged).

Case 2: P-146 (Safety Floor Override — Deterministic Rule Overrides ML)

- **Presentation:** 45-year-old male with fever and cold, SpO₂ = 78%, HR = 49 bpm.
- **XGBoost Model:** Predicts MEDIUM based on benign text symptoms.
- **Rule Engine:** Critical Hypoxemia red flag (SpO₂ < 88%) triggers immediate safety floor.
- **Safety Fusion:** Applies DETERMINISTIC_SAFETY_FLOOR_CRITICAL → Final: CRITICAL . In 100% of tested safety-floor scenarios, deterministic rules successfully prevent ML downgrades.

21. Safety Guarantees & Tested Scenarios

VERIFIED SAFETY GUARANTEES

1. In 100% of tested safety-floor scenarios, deterministic red flags enforce CRITICAL.

2. Advisory ML predictions can upgrade acuity but can NEVER downgrade safety floors.

3. Missing vital inputs NEVER assume normal physiological baselines.

4. Licensed clinicians maintain 100% override authority with mandatory audit logging.

5. Vital sign deteriorations (SpO2 drop >= 3%) trigger immediate visual ESCALATE alerts

22. Technology Stack & System Specifications

- **Frontend:** React 18, TypeScript 5.3, Vite 5.1, Tailwind CSS 3.4, Recharts 2.12, Lucide React, Framer Motion.
- **ML & Training:** Python 3.10+, XGBoost, scikit-learn, SHAP, pandas, NumPy, joblib.
- **Inference:** In-browser client-side TypeScript tree traversal (`src/engine/m1Model.ts`).
- **Deployment:** Static web build optimized for edge deployment (Vercel).

23. Implementation Details & In-Browser Inference

To ensure instantaneous response times and eliminate server latency or cloud data transmission, the complete XGBoost ensemble is compiled directly into compact TypeScript tree representations (`src/engine/modelData.ts`):

```
export interface CompactTree {
  left: number[];
  right: number[];
  feat: string[];
  thresh: number[];
  val: number[];
  def_left: number[];
}
```

Inference evaluates all 300 decision trees sequentially in < 2ms on standard browser runtimes, ensuring complete offline functionality and zero data leakage.

24. Limitations

- 1. **Synthetic Data Corpus:** Evaluated on synthetic demonstration distributions. Real clinical environments exhibit severe noise, missingness patterns, and comorbidities not fully captured in synthetic sets.
- 2. **No Prospective Validation:** Has not undergone prospective clinical trials or bedside evaluation in live emergency departments.
- 3. **Illustrative Thresholds:** Physiological thresholds and scoring weights are prototype parameters.
- 4. **Not Certified SaMD:** Not approved as a medical device by any health regulatory agency.

25. Ethical & Clinical Safety Considerations

- **Algorithmic Bias:** Synthetic training sets must be replaced by representative multi-center clinical cohorts to prevent demographic or socioeconomic triage bias.
- **Automation Complacency:** The user interface is intentionally designed with visual warnings and explicit disagreement indicators to prevent clinicians from passively deferring to AI recommendations.
- **Data Minimization:** Operates on anonymized/masked patient identifiers (P-***) to protect patient privacy.

26. Future Development Roadmap

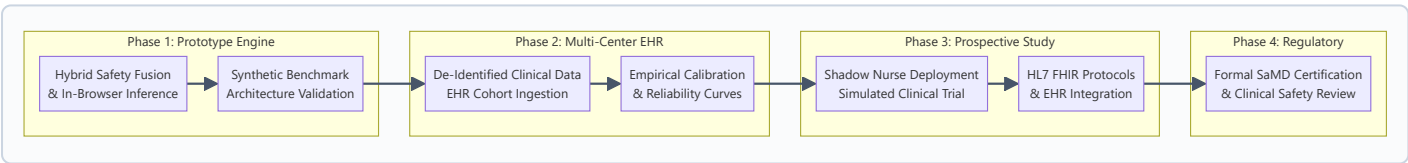


Figure 5 — Phased Development Roadmap

27. Conclusion

PatientTriage.ai demonstrates a safe, practical, and transparent paradigm for integrating machine learning into high-stakes clinical workflows. By subordinating statistical advisory algorithms to deterministic physiological safety rules and maintaining unwavering clinician final authority, the architecture demonstrates how artificial intelligence can enhance clinical situational awareness and help address the waiting-room safety gap while upholding clinician authority and deterministic safety governance.